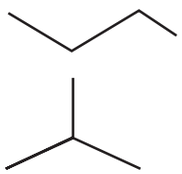
IUPAC Name	Molecular Formula	Condensed Structural formula
Methane	CH_4	CH_4
Ethane	C_2H_6	CH_3-CH_3
Propane	C_3H_8	CH_3-CH_2-CH_3

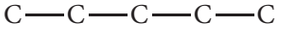
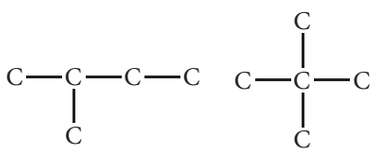
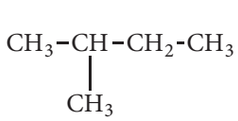
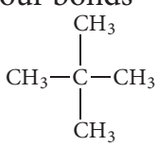
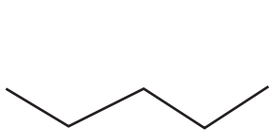
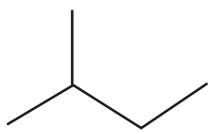
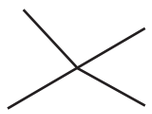
However, higher members can have more than one structure leading to constitutional isomers (differ in connectivity) or structural isomers. For example, an alkane with molecular formula C_4H_{10} can have two structures. They are n-butane and iso-butane. In n-butane, all the four carbon atoms are arranged in a continuous chain. The 'n' in n-butane stand for 'normal' and means that the carbon chain is unbranched. The second isomer iso-butane has a branched carbon chain. The word iso indicates it is an isomer of butane.

Molecular Formula	Structural Formula	Bond line Formula	IUPAC name	Common Name
C_4H_{10}	CH_3-CH_2-CH_2-CH_3		butane	n-butane
	$CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-CH_3$		2-methylpropane	iso-butane

Though both the structures have same molecular formula but their carbon chains differ leading to chain isomerism

Let us understand the chain isomerism by writing the isomers of pentane C_5H_{12}

Solution:

To begin, draw the carbon backbone of the straight chain-isomer	
Determine the carbon backbone structure of the other isomers by arranging the carbon atoms in the other way	
Fill in all the hydrogen atoms so that each carbon forms four bonds CH_3-CH_2-CH_2-CH_2-CH_3  	
  	

Unit 13 - jagan.indd 181 22-01-2021 12:51:25